

Assignment 4: Multivariate Gaussian Density Functions, Covariance Matrices and Constant Density Curves

AI5707 - Pattern Recognition and Machine Learning

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1. Introduction

Data was generated from bivariate Gaussians with three kinds of covariance (isotropic, diagonal and full). For each one I estimated the mean and covariance from the samples, worked out the eigenvalues and eigenvectors, and drew the curves along which $(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)$ stays constant. These curves turn out to be ellipses, and the main thing to check was whether their size and tilt really follow from the eigenvalues and eigenvectors.

2. Experiments performed

Every case uses 1000 points. I started with two independent Gaussian samples, $D1 \sim N(\mu_1, 1)$ and $D2 \sim N(\mu_2, 1)$, generated with the Marsaglia polar method from the first assignment, and set $\mu = (2, 3)$. D1 is the first coordinate and D2 the second, so each point is $\mathbf{x} = (x_1, x_2)$. The covariance is estimated with the usual $1/(N - 1)$ formula. For a 2×2 symmetric matrix the eigenvalues have a closed form, so I used that instead of an iterative method.

The three plots are laid out the same way. Grey dots are the samples and the black cross is the estimated mean. The three closed curves are where $d = 1, 2$ and 3 , with $d^2 = (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)$. The two arrows are the eigenvectors of the estimated covariance. The 2 arrows were made as long as the matching axis of the $d = 1$ curve, which is $2\sqrt{\lambda}$, and the legend on every figure gives the eigenvalue and axis length. Table 1 collects the numbers. Figure 1 shows the isotropic case.

Isotropic covariance (Problem 1)

With $C = \sigma^2 I$ and $\sigma^2 = 1$, the estimated mean came out at $(2.007, 3.041)$ and the estimated covariance was $[[0.928, -0.045], [-0.045, 0.937]]$. The eigenvalues are 0.978 and 0.887, against 1 and 1 for the true matrix. In Figure 1 the curves are almost perfect circles, and the two axis lengths, 1.978 and 1.884, are both close to 2. The arrows point along the diagonals.

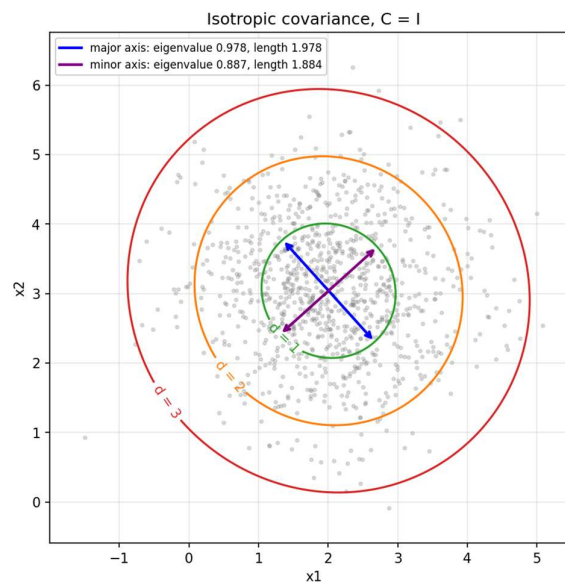


Figure 1. Isotropic case. The curves are close to circles and the eigenvector directions are arbitrary.

Diagonal covariance (Problem 2)

For the diagonal case I took $C = \text{diag}(4, 1)$ and transformed the same points with $Y = \mu + \Sigma^{1/2}(X - \mu)$. Since $\Sigma^{1/2} = \text{diag}(2, 1)$, this stretches the cloud to twice its width along x_1 around the mean and leaves x_2 alone. The estimated covariance is $\begin{bmatrix} 3.712 & -0.090 \\ -0.090 & 0.937 \end{bmatrix}$ with eigenvalues 3.715 and 0.934, against a true 4 and 1. In Figure 2 the ellipses are elongated horizontally. The major axis makes an angle of -1.85° with the x_1 axis, so it is essentially horizontal, and the axis lengths of 3.855 and 1.933 are in a ratio of 1.99, which matches what the square roots of 4 and 1 predict.

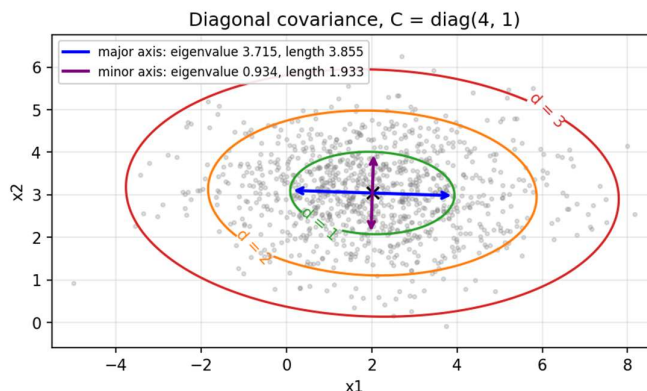


Figure 2. Diagonal case. The axes stay parallel to the coordinate axes.

Full covariance (Problem 3)

For the third case I used $C = \begin{bmatrix} 3 & 1.8 \\ 1.8 & 2 \end{bmatrix}$, which has a correlation of about 0.73 and true eigenvalues 4.368 and 0.632. There is no neat square root to write down by hand here, so I computed $\Sigma^{1/2} = V\Lambda^{1/2}V^T$ from the eigen decomposition and applied the same transformation to the same X . The estimated covariance is $\begin{bmatrix} 2.697 & 1.568 \\ 1.568 & 1.800 \end{bmatrix}$ with eigenvalues 3.879 and 0.617. In Figure 3 the ellipses are tilted, and the major axis sits at 37.02° , very close to the 37.24° the true matrix gives.

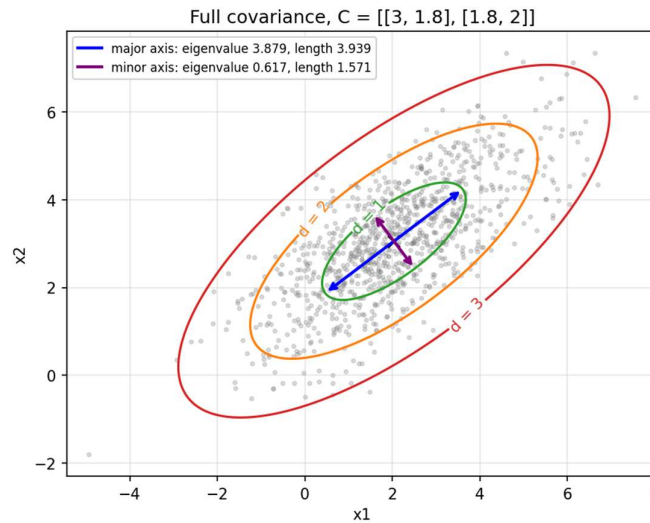


Figure 3. Full covariance case. The off-diagonal term tilts the ellipses by about 37°.

Case	True eigenvalues	Estimated eigenvalues	Axis lengths at $d = 1$ (major / minor)	Estimated major axis angle
Isotropic	1, 1	0.978, 0.887	1.978 / 1.884	not meaningful
Diagonal	4, 1	3.715, 0.934	3.855 / 1.933	-1.85° (true 0°)
Full	4.368, 0.632	3.879, 0.617	3.939 / 1.571	37.02° (true 37.24°)

Table 1. True and estimated eigenvalues for the three covariance matrices.

Two classes and their decision boundary (Problem 4)

For the last problem I generated two classes, ω_1 and ω_2 , with 500 points each from $N(\mu_i, \Sigma_i)$, using the same transformation as before. To draw the boundary I assumed equal priors and gave each point to the class with the larger value of $g_i(x) = -\frac{1}{2}(x - \mu_i)^T \Sigma_i^{-1}(x - \mu_i) - \frac{1}{2} \ln |\Sigma_i|$. The boundary is where $g_1(x) = g_2(x)$. The mean and covariance of each class are estimated from its own samples, and the accuracy in each panel of Figure 4 is measured on those same samples, so it is on the optimistic side. I tried four settings.

In A both classes have covariance I , with means $(0, 0)$ and $(4, 0)$. The boundary is a nearly vertical straight line at $x \approx 2$, halfway between the means and at right angles to the line joining them. It isn't perfectly straight, which I put down to the two estimated covariances differing a little (1.39 and 1.57 on the diagonal for class 2, against 1.51 and 1.53 for class 1). Accuracy is 95.7%.

In B both classes share $C = [[3, 1.8], [1.8, 2]]$ and the means are the same as in A. The boundary is still straight, but now it is tilted, running up and to the right, and it is no longer perpendicular to the line between the means, which is horizontal. It leans the same way the clouds do. Accuracy is 95.6%, almost identical to A, because the distance between the means measured with Σ^{-1} is 3.4 here and 3.3 in A, and that is what decides how much the classes overlap.

In C class 1 has covariance I and class 2 has $[[4, 1.5], [1.5, 2]]$, with means $(0, 0)$ and $(4, 0)$. Now the boundary is curved and wraps around the tighter class 1 cloud. Class 2 is much more spread out, so the two clouds overlap more along the boundary and accuracy falls to 93.5%.

In D both classes have the same mean, $(2, 2)$, but class 1 has covariance $0.5I$ and class 2 has $[[4, 1.5], [1.5, 2]]$. The boundary is a closed, roughly circular curve around the shared mean, with class 1 inside and class 2 everywhere outside. This was the hardest case at 77.3%, since the two clouds sit right on top of each other and only their spread differs.

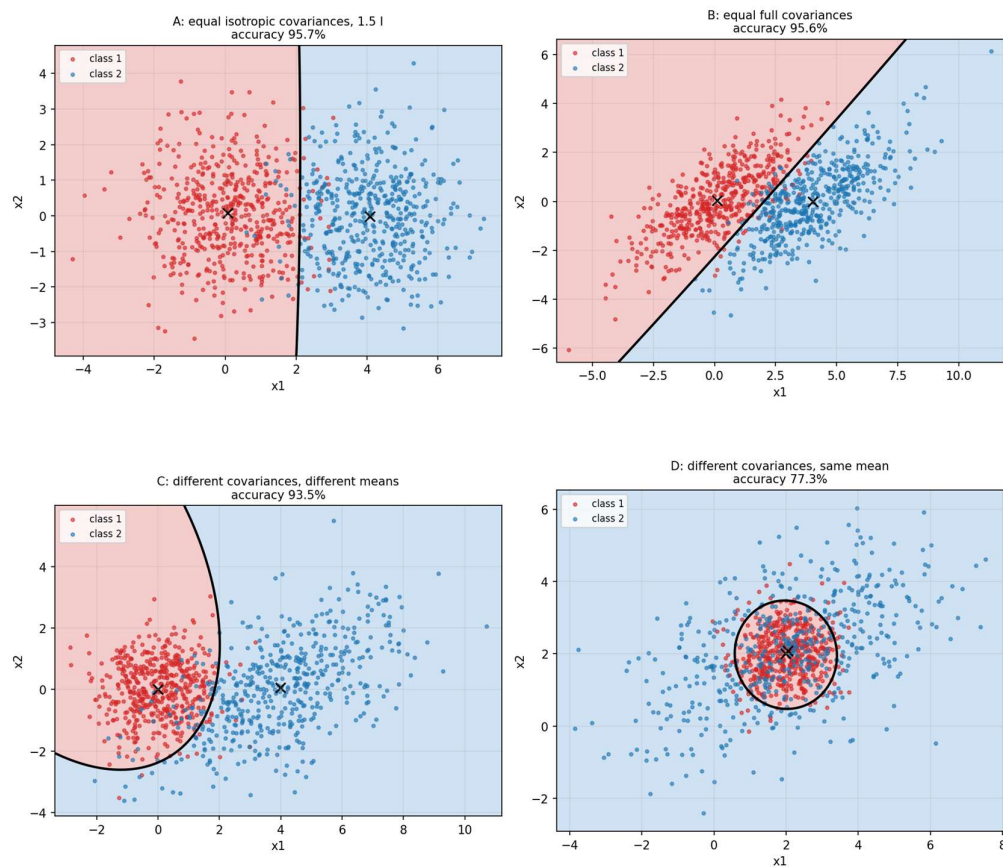


Figure 4. Two-class experiments. Shading shows which class the rule picks, the black curve is the decision boundary and the crosses are the estimated class means.

3. Inference

Once the mean is known, the covariance matrix fixes the shape of the density. The eigenvectors give the directions of the ellipse axes and the eigenvalues give the variance along them, so the axis lengths grow with the square root of the eigenvalue and not with the eigenvalue itself. The diagonal case shows this nicely: variances of 4 and 1 gave axes in a ratio of about 2 to 1.

The off-diagonal entry is what tilts the ellipse. With a diagonal C the axes stay parallel to x_1 and x_2 , and with a non-zero covariance they rotate, which is why the transformation needs the matrix square root and not just the square root of each entry. The isotropic case taught me something different: when the eigenvalues are equal, the eigenvectors carry no information, and the directions I got were decided by noise.

The estimates were close to the true values but not exact with 1000 points. The worst was the major eigenvalue in Problem 3, about 11% low, and that traced back to the starting sample and not to the method. I would expect the gap to shrink with more points, although I haven't tested that.

On the decision boundaries, the covariances decide the shape of the boundary and the means decide where it sits. Equal covariances gave a straight line (A and B). In A, where the covariance was a multiple of the identity, the line was perpendicular to the segment joining the means, and in B it was not. Once the covariances differ, the boundary bends (C), and when the means coincide it can close into a loop around them (D), a case where no straight line could work at all. Accuracy went down as the classes overlapped more, and was lowest when only the spread separated them.